

DeXposure-Agent: An Agentic System for DeFi Risk Monitor and Decision based on Graph Times-Series Foundation Model

Aijie Shu and Fengxiang He

Abstract—Decentralised finance (DeFi) has become a densely interconnected financial system in which credit exposures propagate through tokenised liabilities across protocols, making timely *risk monitoring* and *decision recommendation* essential for investors, protocol operators, and regulators. While recent “agentic” pipelines promise automated surveillance and intervention planning, this direction remains premature in practice, in part because of the lack of a reliable *foundation model backbone* that can forecast the evolving exposure network with calibrated uncertainty and sufficient data efficiency. Building on DeXposure-FM, a graph time-series foundation model trained on over 43.7 million data entries, we present two coupled contributions. First, *DeXposure-Framework*: a four-layer query architecture (prediction → monitoring and scenario → LLM decision → safety gate) that transforms FM-predicted exposure graphs into auditable supervisory tickets. Second, *DeXposure-Bench*: the first standardised evaluation suite for FM+LLM decision pipelines on temporal exposure graphs, comprising six benchmarks with a regulator-aligned ground truth defined by absolute weight loss. Across the suite we show that (1) the FM signal and the LLM decision layer each contribute independently to decision quality (F1 +33% / +27% in layer-wise ablations), (2) prediction intervals are well-calibrated (coverage 0.913 vs the 0.90 target), and (3) the gated stack produces grounded, high-quality recommendations (grounding 1.0, judge 2.69 / 5).

Code: <https://github.com/EVIEHub/DeXposure-Agent>

Demo: <https://www.youtube.com/>

Index Terms—Agentic system, graph, time series, foundation model, decentralised finance, financial network, risk modelling.



1 INTRODUCTION

In May 2022, the Terra/Luna collapse erased over \$40 billion in value within 48 hours, triggering cascading liquidations across lending protocols, DEXs, and yield aggregators that held UST-denominated assets. Five months later, the FTX insolvency exposed hidden cross-protocol exposures and wiped out another \$8 billion. These crises share a common pattern: shocks propagated through an interconnected web of tokenised liabilities—credit exposures—that no single participant fully observed. In decentralised finance, exposures are high-dimensional, time-varying, and often implicit; data are noisy and incomplete; and crises unfold within days, leaving limited time for manual assessment.

Automated *agentic* pipelines offer a principled response: an agent can ingest on-chain measurements, maintain a belief over the evolving exposure network, surface early-warning signals, stress-test plausible shocks, and recommend risk-mitigating actions with auditable rationale. However, existing agentic proposals for DeFi risk largely rely on general-purpose large language models or static heuristics, and lack a dedicated forecasting backbone for graph-structured time series. Without accurate and data-efficient prediction of network dynamics, monitoring agents suffer from high false-alarm rates and delayed detection, while decision agents risk recommending actions based on brittle or poorly calibrated beliefs. In high-stakes contexts, uncer-

tainty calibration and robustness to missingness and regime shifts are not optional—they determine whether an agent is trusted and deployable.

This paper addresses that bottleneck by building an agentic system around a strong graph time-series foundation model, and—in tandem—by providing the evaluation suite the community needs to compare such systems. We leverage DEXPOSURE-FM, a foundation model trained on over 43.7 million data entries from DeFi exposure networks, designed to forecast the evolution of credit-exposure graphs across multiple horizons. On top of this backbone we present DEXPOSURE-FRAMEWORK, a modular four-layer query architecture that follows a *forecast–measure–decide* paradigm, and DEXPOSURE-BENCH, a standardised evaluation suite that benchmarks FM+LLM decision pipelines on a shared regulator-aligned ground truth.

Our contributions are:

- 1) **DEXPOSURE-FRAMEWORK: a four-layer query architecture for DeFi exposure-network risk (§4)**. We design and implement an end-to-end system organised as four layers—FM prediction, deterministic monitoring and scenario engines, LLM-based decision, and a safety gate—that transforms forecasted exposure graphs into auditable supervisory tickets. Every component is deterministic and reproducible: five network risk metrics with rolling baselines, five standardised stress scenarios with CVaR aggregation, and a four-level playbook governed by data-health and confidence gates. While prior agentic work in DeFi targets transaction verification [1] or intent mining [2], and graph-based foundation models for risk

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have been explored only in traditional finance [3], no existing system unifies these capabilities for DeFi exposure networks.

- 2) **DEXPOSURE-BENCH: an evaluation suite for FM+LLM decision pipelines** (§5). We release the first standardised harness that scores a candidate pipeline on six axes—forecasting quality, streaming anomaly detection, predictive uncertainty, what-if scenario fidelity, supervisory decision quality, and data-quality robustness—on the 16,610-protocol DEXPOSURE dataset. The suite introduces a regulator-aligned ground truth defined by *absolute* weight loss rather than fractional change, which restores discrimination between methods that target central versus peripheral nodes. Generic LLM-agent benchmarks (SWE-BENCH, AGENTBENCH, HELM) do not evaluate domain-grounded decisions, and temporal-graph benchmarks (TGB, OGB) stop at structural prediction; DEXPOSURE-BENCH fills the niche.
- 3) **Layer-wise ablations and calibration evidence** (§6). On DEXPOSURE-BENCH we show that the FM prediction layer and the LLM decision layer each contribute independently to ticket F1 (+33% and +27% respectively in controlled ablations), the safety gate raises LLM-as-judge explanation quality from 2.48 to 2.69 / 5, and FM prediction intervals are well-calibrated (coverage 0.913–0.919 vs the 0.90 target). Components that appear inactive under clean test conditions (data-health gating, multi-horizon diversity) are shown to be safety reserves that activate under data degradation and crisis dynamics.

An open-source implementation is available at <https://github.com/EVIEHub/DeXposure-Agent>.

2 RELATED WORKS

This section surveys prior work most closely related to our setting.

2.1 Financial Network Risk Modelling

In traditional finance, credit risk modelling typically spans both entity-level default modelling and system-level network perspectives. Classical structural and reduced-form approaches focus on estimating default likelihoods for individual institutions, whereas network contagion models study how distress can cascade through interconnected balance sheets. For instance, Dolfin et al. use network epidemic models to study credit contagion, showing that dense interconnections and overlapping portfolios can magnify systemic credit risk [4].

In decentralised finance, credit risk takes a different form because lending is commonly over-collateralised and enforced by smart contracts. Bertomeu et al. introduce parsimonious aggregate risk measures for DeFi lending protocols based only on total deposits and borrowings, and find that systemic fragility rose sharply around mid-2021 during periods of crypto-market stress [5]. Bridging conceptual links between TradFi and DeFi, Aufiero et al. review mechanisms of systemic risk and argue that, while familiar ingredients (leverage, liquidity shocks, correlated exposures) exist in both systems, DeFi’s algorithmic execution and composability introduce distinctive channels through which shocks may propagate [6].

2.2 Machine Learning Approaches

Machine learning methods are increasingly used for systemic-risk detection and contagion analysis [7], [8]. In particular, graph neural networks (GNNs) exploit relational structure to strengthen predictive performance on networked risk problems [9], [10]. Balmaseda et al. demonstrate that GNN-based models can substantially outperform conventional ML when classifying the systemic importance of banks in simulated financial networks [7]. Gonon et al. combine GNNs with explicit interbank liability networks to compute systemic risk measures defined via the minimum capital required to stabilise the system [8]. Their formulation effectively captures contagion channels (e.g., Eisenberg–Noe-style clearing) by embedding graph-structured information directly into the risk aggregation procedure.

2.3 Foundation Models

Foundation models are increasingly influencing finance and economics by shifting from narrowly tuned, task-specific models towards reusable representations learned from broad data. Rather than training a bespoke model per downstream objective, recent work pretrains large neural networks—predominantly transformer-based architectures [11] and related variants—on massive corpora, then transfers these models across tasks via prompting or lightweight adaptation [12], [13], [14]. Alongside this, an emerging policy-oriented literature uses large language models to extract quantitative signals from narrative sources (e.g., reports, releases, and news) and to integrate them into nowcasting pipelines, suggesting that relatively small yet information-rich text inputs can improve real-time forecasting when paired with standard indicators [15], [16], [17].

Foundation-model ideas are also being extended to graph-structured inputs by augmenting self-attention with graph-aware structural encodings, illustrating the viability of large pretrained models for networks that resemble webs of financial contracts or cross-asset relationships; Graphormer provides a representative example [18]. For tabular settings, the Graph Prior-data Fitted Network (GraphPFN) [19] is pretrained on millions of synthetic tabular tasks and reports strong out-of-the-box performance across a range of benchmarks.

2.4 Agentic Systems in Finance

The recent proliferation of large language model (LLM) agents has prompted several applications in finance and DeFi security. In portfolio management, Luo et al. propose a multi-agent LLM framework for automated crypto portfolio construction with team-level collaboration, though without graph-structured market models [20]. BlindTrade [21] combines GNN encoders with four LLM agents for stock portfolio optimisation, representing the closest integration of graph models and LLM agents in finance; however, it targets equity markets and does not address systemic risk in DeFi networks.

Within DeFi, agentic pipelines have been developed primarily for security auditing and transaction analysis. Arbiter [1] employs a Graph-of-Thoughts reasoning framework to verify intent–transaction alignment in DeFi, while

Know Your Intent [2] uses a multi-perspective LLM agent to mine user transaction intents. On the security side, Heimdallr [22] and Knowdit [23] apply agentic pipelines for smart contract vulnerability detection, and PMDetector [24] uses static analysis augmented by LLM reasoning to identify DeFi price manipulation vulnerabilities.

Notably, none of these systems combine a dedicated graph time-series foundation model with an agentic decision pipeline for *systemic risk monitoring* in DeFi exposure networks. Watson et al. [25] use GCN anomaly detection with LLM-generated explanations for Bitcoin fraud, but their pipeline is not agentic (human-in-the-loop) and targets individual transactions rather than network-level risk. In traditional finance, Li et al. [3] pretrain a heterogeneous graph autoencoder for bond default risk propagation, demonstrating graph pretraining for risk but without agent or DeFi components. DeXposure-Agent bridges this gap by integrating a graph time-series foundation model backbone with a structured agent pipeline for DeFi exposure-network risk.

3 NOTATION AND PRELIMINARIES

This section introduces the key terminology and notation used throughout the paper.

3.1 Credit Exposure in DeFi Networks

Let $X_G(q)$ be the set of tokens issued (or generated) by protocol q , and let $X_p(t)$ be the set of tokens held (or locked) by protocol p at time t . We say that protocol p has *credit exposure* to protocol q at time t if p holds at least one token issued by q :

$$X_p(t) \cap X_G(q) \neq \emptyset. \quad (1)$$

Equivalently, p is exposed to liabilities of q : if tokens issued by q devalue or become illiquid, p incurs losses. This token-mediated dependence generalises counterparty credit risk from traditional finance to DeFi, where exposures emerge through tokenised claims rather than explicit bilateral contracts. In practice, these dynamics can be tracked by reconstructing protocol-level TVL time series and monitoring how token composition changes over time [26].

Figure 1 shows how a single workflow can generate layered exposures. A user deposits ETH into Lido and receives stETH; stETH is wrapped into wstETH; wstETH is then deposited into Pendle, which splits it into PT-wstETH and YT-wstETH. On balance sheets, each step turns the upstream token into the downstream protocol's asset, so exposures stack: Pendle is exposed to the wstETH wrapper, which is exposed to Lido. If Lido suffers slashing or stETH de-pegs from ETH, losses can propagate through wstETH into Pendle and its PT/YT holders. This illustrates why DeFi exposures are naturally multi-layered and why DeXposure aims to measure them.

3.2 Mapping Tokens to Issuing Protocols

To implement Eq. (1), we must identify the issuing (or managing) protocol for each token. Let $\mathcal{P}$ be the set of protocols and $\mathcal{C}$ the set of blockchains. For any token σ appearing in protocol p , a *mapping function* $M(\sigma)$ returns

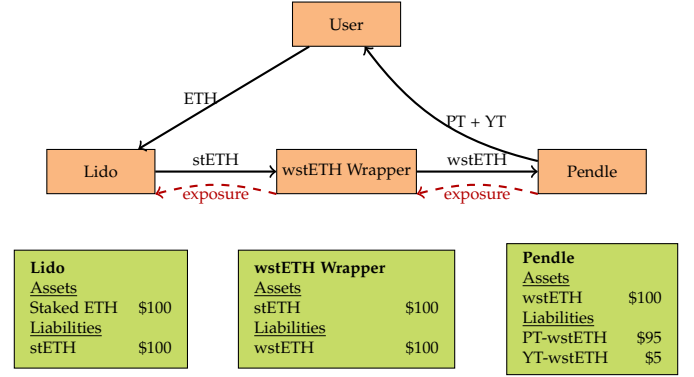


Fig. 1. Multi-layer credit exposure in DeFi. A user stakes ETH with Lido, receiving stETH, which is wrapped into wstETH and deposited into Pendle. Each protocol holds the liability of the preceding one, creating a chain of credit exposures from Pendle to Lido.

the protocol q that issues or governs σ . We construct M via the following fallback pipeline:

- 1) *Metadata lookup*: match tokens to issuing protocols using DefiLlama metadata when available [27].
- 2) *Manual mapping*: for high-TVL tokens without metadata links, assign issuers using documentation and expert judgment.
- 3) *Text-vector similarity*: otherwise, vectorise token and protocol descriptions using TF-IDF [28] and assign $M(\sigma)$ as the most similar protocol, provided the cosine similarity exceeds a threshold θ .
- 4) *Primary market tokens*: if no match is obtained, treat the token as its own protocol (e.g., WETH).

This procedure attributes token movements to the appropriate issuer even when tokens are bridged or wrapped, and is used in curating DeXposure.

3.3 Network Representation and Value Flows

Over a discrete interval $\tau = t_2 - t_1$, we represent exposures by a weighted directed graph $G_\tau = (P_\tau, E_\tau)$, where P_τ is the set of active protocols and E_τ the directed edge set. Each node $p \in P_\tau$ receives a *node weight* equal to the value of its held assets at t_2 :

$$w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2, \sigma}, \quad (2)$$

where $v_{t_2, \sigma}$ is the USD value of token σ at time t_2 . Nodes below a threshold θ may be pruned.

To define the *edge weight* from p to q , we first compute token-level value flows:

$$F_{pq, \tau}^\sigma = \begin{cases} \max(0, -\Delta S_{p, \tau}^\sigma), & \text{if } \Delta S_{p, \tau}^\sigma < 0, \\ \max(0, \Delta S_{q, \tau}^\sigma), & \text{if } \Delta S_{q, \tau}^\sigma \geq 0, \end{cases} \quad (3)$$

where $\Delta S_{p, \tau}^\sigma = v_{p, t_2}^\sigma - v_{p, t_1}^\sigma$ is the change in the USD value of token σ held by p over τ . Intuitively, if p reduces holdings while q increases holdings, σ is interpreted as flowing from p to q ; the construction enforces non-negativity.

The *edge weight* $w_\tau(e_{pq})$ aggregates flows over tokens mapped to issuer q :

$$w_\tau(e_{pq}) = \sum_{\sigma \in X_{pq, \tau}} F_{pq, \tau}^\sigma, \quad (4)$$

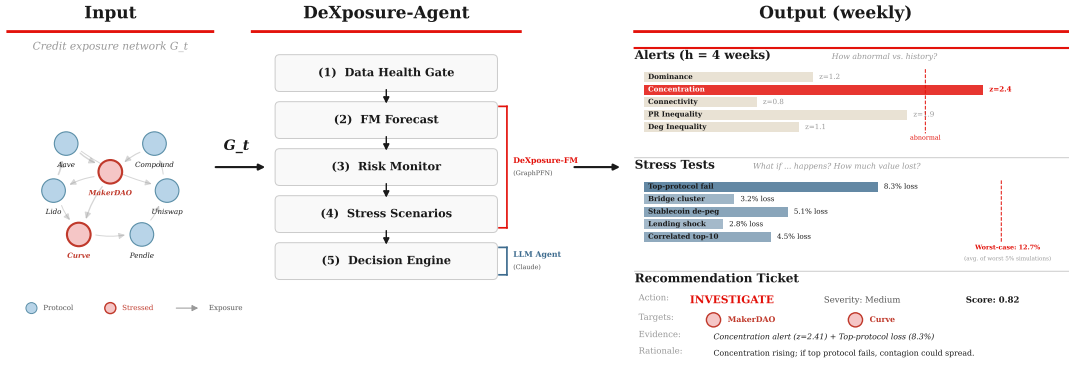


Fig. 2. System overview of DeXposure-Agent. The agent ingests weekly exposure snapshots, queries the DeXposure-FM backbone for multi-horizon forecasts, computes risk metrics and stress scenarios, and emits auditable recommendation tickets under data-health and confidence gates.

where $X_{pq,\tau}$ is the set of tokens issued by q that move from p to q over τ . Positive $w_\tau(e_{pq})$ corresponds to increasing exposure from p to q ; for single-token protocols, that token's flow equals the edge weight. In summary, we compute G_τ by aggregating holdings at t_1 and t_2 , assigning node weights, pruning small nodes, and summing positive token flows into issuer-specific edge weights. The resulting graph sequence captures the evolving credit-dependency structure curated in DeXposure.

4 ALGORITHM

This section specifies the agentic system built on DEXPOSURE-FM for two tasks: network risk monitoring (Section 4.3–4.4) and decision recommendation (Section 4.5).

Figure 2 provides an architectural overview of the full system.

4.1 System Interface

At each decision epoch t , the agent observes a credit-exposure snapshot $G_t = (V, E_t, W_t)$ and node covariates X_t , and interacts with DEXPOSURE-FM through a single forecast primitive:

$$\mathcal{P}_{t,h} \leftarrow \text{DEXPOSURE-FM.Forecast}(G_t, X_t, h), \quad \forall h \in \mathcal{H}, \quad (5)$$

where $\mathcal{P}_{t,h}$ denotes the model's predictive distribution over the future network G_{t+h} , and $\mathcal{H} = \{1, 4, 8, 12\}$ (weeks). From $\mathcal{P}_{t,h}$ we construct a representative predicted graph $\hat{G}_{t+h}$ using a thresholded expected-weight operator:

$$\hat{w}_{pq,t+h} := \Pr(e_{pq,t+h} = 1) \cdot \mathbb{E}[w_{pq,t+h} \mid e_{pq,t+h} = 1], \quad (6)$$

$$\hat{E}_{t+h} := \{(p, q) : \Pr(e_{pq,t+h} = 1) > \pi_{\min}\}. \quad (7)$$

where $\pi_{\min}$ (default 0.2, tuned on validation) balances edge coverage and sparsity. For uncertainty quantification, we draw M Monte Carlo samples $\{\tilde{G}_{t+h}^{(m)}\}_{m=1}^M$ by applying Bernoulli edge sampling and Gaussian weight perturbation to $\mathcal{P}_{t,h}$ (default $M = 50$).

The agent returns three outputs: (a) a set of monitoring alerts with evidence and confidence, (b) a scenario stress summary, and (c) a ranked list of recommendation tickets.

4.2 Data-Health Gating

Real-world DeFi graphs can be sparse, stale, or corrupted. Before producing actionable outputs, the agent computes a scalar data-health score from four deterministic checks—freshness, feature missingness, topology sanity (edge-to-node ratio), and discontinuity detection:

$$DH_t := \text{mean}(\text{fresh}_t, \text{miss}_t, \text{topo}_t, \text{disc}_t) \in [0, 1],$$

$$\text{SAFE_MODE}_t := \mathbb{I}\{DH_t < \tau_{\text{data}}\}. \quad (8)$$

When DH_t falls below τ_{data} (default 0.7), the system enters safe mode: monitoring alerts are still emitted, but intervention-type recommendations are suppressed (Section 4.5).

4.3 Task I: Network Risk Monitoring

Monitoring metrics (Φ): Risk in exposure networks manifests as concentration (a few protocols dominate), fragility (shocks propagate widely), and instability (structure shifts rapidly). To capture these dimensions, the monitoring agent evaluates five functionals on $\hat{G}_{t+h}$: Formally, at each horizon h we compute:

$$\mathbf{M}_{t,h} := \Phi(\hat{G}_{t+h}, X_t) \in \mathbb{R}^5, \quad \tilde{\mathbf{M}}_{t,h}^{(m)} := \Phi(\tilde{G}_{t+h}^{(m)}, X_t). \quad (9)$$

Alert generation.: An alert fires when a predicted metric departs from its rolling historical baseline by more than z standard deviations (default $z = 2$, baseline window $L = 26$ weeks):

$$\mathcal{A}_t := \left\{ a = (h, j) : \frac{|M_{t,h}^{(j)} - \mu_t^{(j)}|}{\max(\sigma_t^{(j)}, \epsilon)} > z \right\}, \quad (10)$$

where $(\mu_t^{(j)}, \sigma_t^{(j)})$ are the rolling mean and standard deviation of metric j over the last L observed snapshots, and $\epsilon = 10^{-8}$ prevents division by zero.

Evidence and attribution.: Each alert $a \in \mathcal{A}_t$ carries an evidence bundle $\text{Ev}_t(a) = (M_{t,h}^{(j)}, \mu_t^{(j)}, \sigma_t^{(j)}, \text{Attr}_t(a), h)$, where $\text{Attr}_t(a)$ is a ranked list of top- K contributing nodes/edges obtained via marginal contribution approximation: $\text{Contrib}(e) := \Phi^{(j)}(\hat{G}_{t+h}) - \Phi^{(j)}(\hat{G}_{t+h} \setminus e)$ (default $K = 10$).

ID	Metric	Definition
M1	Systemic Importance	$\max_{p \in V} \text{PageRank}(\hat{G}_{t+h})_p$
M3	HHI Concentration	$\sum_p (d_p / \sum_q d_q)^2$, d_p = weighted out-degree
M4	Network Density	$ E_{t+h} / V (V - 1)$
M6	PageRank Gini	$\text{Gini}(\{\text{PR}_p\}_{p \in V})$
M7	Degree Gini	$\text{Gini}(\{d_p\}_{p \in V})$

TABLE 1

Monitoring functionals Φ . PageRank uses power iteration with damping 0.85 and 10 iterations. Gini coefficient: $1 - 2 \sum_{i=1}^n \sum_{k=1}^i x_{(k)} / (n \cdot \sum_k x_{(k)})$, where $x_{(k)}$ are sorted values.

ID	Scenario	Target	Shock
S1	Single protocol failure	Highest-weight node	100%
S2	Bridge cluster failure	All bridge-category nodes	100%
S3	Stablecoin de-peg	All stablecoin nodes	50%
S4	Sector-wide shock	All lending nodes	30%
S5	Correlated stress	Top-10 nodes by weight	20%

TABLE 2

Standardised stress-scenario library $\mathcal{S}_0$.

Action	Severity	Weight w_u	Safe-mode OK?	Min. alerts
Monitor	Low	0.25	Yes	1
Investigate	Medium	0.50	Yes	1
Recommend	High	0.75	No	2
-Reduce				
Contingency	Critical	1.00	No	3

TABLE 3

Playbook action space $\mathcal{U}$ with severity weights and feasibility constraints.

Uncertainty-aware confidence.: Alert reliability depends on three factors: data quality, forecast dispersion, and horizon length. Using Monte Carlo samples, we compute the coefficient of variation $\text{CV}_{t,h}^{(j)} := \sigma(\{\tilde{M}_{t,h}^{(m,j)}\}_m) / |\tilde{M}_{t,h}^{(j)}|$ (clamped to $[0, 1]$) and define:

$$C_t(a) := DH_t \cdot \frac{1}{1 + \text{CV}_{t,h}^{(j)}} \cdot \frac{1}{1 + \ln(1 + h)} \in [0, 1]. \quad (11)$$

This score monotonically decreases with lower data health, wider predictive spread, and longer horizons, ensuring that unreliable alerts are automatically downweighted.

4.4 Scenario Engine

To quantify tail risk beyond metric-level alerts, the agent evaluates a standardised stress-scenario library $\mathcal{S}_0$. Each scenario applies a deterministic shock—removing or reducing edge weights of targeted nodes—and propagates losses through the network:

For each scenario s and horizon h , we compute the systemic loss on both the predicted graph and each MC sample:

$$L_{t,h}(s) := \frac{\sum_e w_e(\hat{G}_{t+h}) - \sum_e w_e(\hat{G}_{t+h}^s)}{\sum_e w_e(\hat{G}_{t+h})}, \quad (12)$$

$$\tilde{L}_{t,h}^{(m)}(s) := \ell(\tilde{G}_{t+h}^{(m)}; s). \quad (13)$$

where $\hat{G}_{t+h}^s$ denotes the network after applying shock s . The scenario engine also tracks the number of *distressed* nodes (those losing $> 50\%$ of total edge weight) and a propagation depth estimate $\lceil \log_2(\text{affected nodes} + 1) \rceil + 1$.

CVaR aggregation.: To capture tail risk, we compute an empirical CVaR from the MC sample losses. Let $\lambda \in [0, 1]$ (default 0.5) control the tail fraction. We sort the M sample losses in descending order and average the worst $\lceil \lambda M \rceil$:

$$\text{CVaR}_\lambda(s, h) := \frac{1}{\lceil \lambda M \rceil} \sum_{i=1}^{\lceil \lambda M \rceil} \tilde{L}_{t,h}^{(i)}(s), \quad (14)$$

where $\tilde{L}^{(1)} \geq \tilde{L}^{(2)} \geq \dots$ are order statistics. The scenario summary $\mathcal{R}_t$ ranks scenarios by $L_{t,h}(s)$ and retains the worst-case scenario across all horizons.

4.5 Task II: Decision Agent

Action space.: The decision module maps alerts and scenario results to recommendation tickets using a conservative playbook with four action types, ordered by escalation severity:

Feasibility constraints.: An intervention-type action u (those requiring safe-mode off) is feasible only when both the data-health gate and the confidence gate are satisfied:

$$u \in \mathcal{C}_t \iff (\text{SAFE_MODE}_t = 0) \wedge (\bar{C}_t \geq \tau_{\text{conf}}), \quad (15)$$

where $\bar{C}_t = \frac{1}{|\mathcal{A}_t|} \sum_{a \in \mathcal{A}_t} C_t(a)$ is the mean alert confidence and τ_{conf} (default 0.6) gates interventions. Non-interventional actions (Monitor, Investigate) are always feasible when at least one alert exists.

Ticket scoring.: Each feasible recommendation is scored by a multiplicative heuristic combining playbook severity, alert confidence, and scenario impact:

$$\text{Score}_t(u) := w_u \cdot \bar{C}_t \cdot \text{Imp}_t, \quad (16)$$

where w_u is the severity weight (Table 3), $\bar{C}_t$ is the mean alert confidence, and $\text{Imp}_t := \frac{1}{|\mathcal{S}_0|} \sum_{s \in \mathcal{S}_0} \text{CVaR}_\lambda(s)$ aggregates scenario impact as the mean CVaR across all scenarios. When no scenarios are available, Imp_t defaults to 1.0 (neutral multiplier), ensuring alert-only tickets receive non-zero scores. The agent returns the top- N tickets (default $N = 10$) ranked by $\text{Score}_t(u)$.

Target selection.: For intervention tickets, targets are selected by collecting the top- K attributed nodes from all triggering alerts and worst-case scenarios, ranked by frequency-weighted vote.

4.6 End-to-End Procedure

Algorithm 1 summarises the full agent loop.

Algorithm 1 DeXposure-Agent: one epoch

Require: State (G_t, X_t) ; horizons $\mathcal{H}$; scenarios $\mathcal{S}_0$; thresholds $(\tau_{\text{data}}, \tau_{\text{conf}}, z, \pi_{\min})$; MC count M

Ensure: Alerts $\mathcal{A}_t$; scenario summary $\mathcal{R}_t$; tickets $\mathcal{D}_t$

- 1: $DH_t \leftarrow \text{DataHealth}(G_t, X_t)$; $\text{SAFE_MODE}_t \leftarrow \mathbb{I}\{DH_t < \tau_{\text{data}}\}$
- 2: **for all** $h \in \mathcal{H}$ **do**
- 3: $\mathcal{P}_{t,h} \leftarrow \text{DEXPOSURE-FM.Forecast}(G_t, X_t, h)$
- 4: $\hat{G}_{t+h} \leftarrow \text{BuildPredGraph}(\mathcal{P}_{t,h}; \pi_{\min})$ $\triangleright$ Eq. (6)
- 5: $\{\tilde{G}_{t+h}^{(m)}\}_{m=1}^M \leftarrow \text{MCSample}(\mathcal{P}_{t,h})$ $\triangleright$ Bernoulli edges + Gaussian weights
- 6: $M_{t,h} \leftarrow \Phi(\hat{G}_{t+h}, X_t)$ $\triangleright$ Table 1: M1,M3,M4,M6,M7
- 7: $\mathcal{A}_{t,h} \leftarrow \{(h, j) : |M_{t,h}^{(j)} - \mu_t^{(j)}| / \sigma_t^{(j)} > z\}$
- 8: Attach $\text{Ev}_t(a)$ and $\hat{C}_t(a)$ for all $a \in \mathcal{A}_{t,h}$ $\triangleright$ Eq. (11)
- 9: **for all** $s \in \mathcal{S}_0$ **do**
- 10: $L_{t,h}(s) \leftarrow \text{StressLoss}(\hat{G}_{t+h}, s)$; compute CVaR from $\{\tilde{G}_{t+h}^{(m)}\}$
- 11: **end for**
- 12: **end for**
- 13: $\mathcal{A}_t \leftarrow \bigcup_{h \in \mathcal{H}} \mathcal{A}_{t,h}$
- 14: $\mathcal{R}_t \leftarrow \text{RankScenarios}(\{L_{t,h}(s), \text{CVaR}_\lambda(s, h)\}_{h,s})$
- 15: $\mathcal{D}_t \leftarrow \text{PlaybookDecision}(\mathcal{A}_t, \mathcal{R}_t, DH_t, \text{SAFE_MODE}_t, \tau_{\text{conf}})$ $\triangleright$ Table 3
- 16: **return** $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$

5 DEXPOSURE-BENCH: AN EVALUATION SUITE FOR FM+LLM DECISION PIPELINES

5.1 Motivation: an evaluation gap

Foundation-model pipelines that pair a domain forecaster with a large language model decision layer are proliferating across finance, supply-chain risk, and infrastructure monitoring. Despite this, the community lacks a standardised suite for evaluating the full *pipeline*—most existing benchmarks evaluate a single stage in isolation. Generic LLM-agent benchmarks (SWE-BENCH [29], AGENTBENCH [30], HELM [31]) cover open-ended coding or tool use but not domain-grounded supervisory decisions. Temporal-graph benchmarks (TGB [32], OGB [33]) measure structural prediction quality but stop short of any decision metric. Domain-specific evaluations of DeFi risk forecasting exist as one-off experiments accompanying single papers, none of them packaged as a reusable harness.

DEXPOSURE-BENCH fills this gap with a single, reproducible suite that scores a candidate pipeline on six axes spanning forecast quality, anomaly detection, predictive uncertainty, what-if query fidelity, downstream decision quality, and data-quality robustness, on a shared 16,610-protocol DeFi exposure dataset with a *regulator-aligned* ground truth.

5.2 Suite schema

The suite is organised as six benchmarks, each producing an independently reportable scalar (or vector of scalars). Pipelines may report against any subset; the full leaderboard requires all six. Table 4 summarises each axis.

The six axes were chosen so that they are: (i) *non-overlapping* in what they measure—a system can excel at b1 while failing b5; (ii) *decomposable along the four-layer framework* (§4), so that ablations on the predictor layer (b1, b3, b6),

monitor layer (b2), scenario layer (b4), and decision layer (b5) can be reported independently; and (iii) *individually publishable*, so that downstream work may extend a single axis without re-running the whole suite.

5.3 Dataset and split

The benchmark draws on the DEXPOSURE weekly exposure-graph dataset [26], a directed weighted graph $G_t = (V_t, E_t, w_t)$ at each weekly timestep t , with V_t a subset of the 16,610-protocol universe observed over 2020-03–2025-08 (283 weeks). Node features carry log-TVL, number of token types, maximum token share, Shannon entropy of the token distribution, and a 15-class category one-hot. Edge weights are log-scaled USD inter-protocol exposures. We adopt the strict no-leakage temporal split of Table 5.

5.4 Ground truth: regulator-aligned absolute loss

Prior systemic-risk evaluations rank protocols by *fractional* weight change, which produces a ground-truth pool dominated by small protocols whose deterioration carries low systemic significance. This choice silently biases every downstream metric and is not what supervisory authorities monitor in practice.

DEXPOSURE-BENCH therefore defines its ground truth in terms of *absolute* exposure loss. Let $w_t(v) = \sum_{e \ni v} w_t(e)$ be the total weight incident on protocol v at time t . For horizon h and lookahead window W we define the *stressed set* $\mathcal{S}_t^h$ as the top π fraction of active protocols ranked by absolute loss:

$$\Delta_t^h(v) = w_t(v) - w_{t+h}(v), \quad \mathcal{S}_t^h = \text{top}_\pi \{v : w_t(v) > 0, \Delta_t^h(v) > 0\}. \quad (17)$$

We fix $\pi = 0.05$ throughout (yielding an average pool of $|\mathcal{S}_t^h| \approx 283$ protocols per week, stable across the test split) and the lookahead window $W = 4$ weeks.

Equation 17 is the single ground-truth definition used by b2, b5, and the LLM-decision evaluation pipeline. Aligning the metric with absolute economic magnitude—rather than fractional change—restores discrimination between methods that target central versus peripheral nodes and matches how a supervisor would prioritise.

5.5 Methods supported out of the box

Nine reference methods are bundled with the harness, sufficient to populate a complete leaderboard from a fresh checkout. Each method exposes the same `predict_graph` or `decide_actions` interface, so adding a tenth method is a single-file change.

The seven listed in Table 6 together expose all four cuts of the framework ablation: (i) adding the predictor layer (m1→m5), (ii) swapping the decision layer (m5↔m6), (iii) adding the safety gate (m6↔m7), and (iv) removing the predictor from the LLM stack (m2↔m6).

5.6 Reporting protocol

To enter the leaderboard, a submission produces one JSON file per {benchmark, method} pair following the harness’s published schema, and a single `results.json` aggregating the suite. Required reports for the headline table are

TABLE 4
DEXPOSURE-BENCH schema. Each row is an independently reportable benchmark; the full leaderboard requires all six.

ID	Capability tested	Primary metrics	Applicable methods
b1_forecast	Temporal graph prediction quality	PageRank MAE, HHI MAE, Gini MAE, trend consistency, rank correlation; per horizon $h \in \{1, 4, 8, 12\}$	Any predictor
b2_warning	Streaming anomaly detection	Precision, recall, F1, lead time, stability; per event $\times$ alert budget	Any monitor + predictor
b3_calibration	Predictive uncertainty quality	PI coverage @ 0.90 target, PI width, ECE, CRPS	Predictors with intervals
b4_stress	What-if scenario fidelity	Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@ k ; per scenario $S_1 \dots S_5$	Predictor + scenario engine
b5_decision	Supervisory ticket quality	Precision, recall@ k , F1, target stability, judge score, FIR	Predictor + decision policy
b6_robustness	Data quality sensitivity	Per-benchmark metrics under 5 degradation regimes; relative degradation	Any predictor

TABLE 5
DEXPOSURE-BENCH canonical split. All reported numbers in this paper come from the test split.

Split	Period	Weeks	Usage
Train	2020-03–2024-06	~220	Predictor training
Validation	2024-07–2024-12	~26	Conformal calibration, threshold tuning
Test	2025-01–2025-08	~33	Frozen leaderboard

TABLE 6
Reference methods bundled with DEXPOSURE-BENCH. Heuristic identifiers use the h namespace; learned methods use m .

ID	Predictor	Policy	Role
h1_weighted_degree	—	heuristic alert	Streaming anomaly base-line
m1_persistence_rules	$G_{t+h} = G_t$	rule engine	Decision base-line
m2_snapshot_llm	current snap	LLM	LLM-only ablation
m3_evolvegcnn	EvolveGCN	—	Learned-GNN base-line
m4_fm_only	FM	—	FM-only ablation
m5_fm_rules	FM	rule engine	FM + rules
m6_fm_llm	FM	LLM	Full FM + LLM stack
m7_fm_llm_gated	FM	LLM + safety gate	Safety-gated FM + LLM

b1, b3, b5, and b6; b2 and b4 are required only for systems that implement a monitor or scenario layer respectively. All metrics are computed on the frozen test split of Table 5; training and hyperparameter tuning may use train + validation freely. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files so that any reported number can be reproduced bit-

TABLE 7
DEXPOSURE-BENCH versus closest existing suites. A check mark indicates a covered axis; — indicates the axis is out of scope.

Suite	Pred.	Anom.	Calib.	Scen.	Decision
OGB [33]	✓	—	—	—	—
TGB [32]	✓	—	—	—	—
HELM [31]	—	—	—	—	✓
SWE-BENCH [29]	—	—	—	—	✓
AGENTBENCH [30]	—	—	—	—	✓
DEXPOSURE-BENCH (this work)	✓	✓	✓	✓	✓

exactly.

5.7 Positioning in the benchmark landscape

Table 7 compares DEXPOSURE-BENCH against the closest existing evaluation suites. No prior suite covers a domain-grounded decision metric on top of a forecasting backbone, which is the niche DEXPOSURE-BENCH addresses.

5.8 Reproducibility and extensibility

The harness, dataset, and reference implementations are released under an open licence at the project repository. A complete leaderboard run, including all nine reference methods and the LLM evaluation pipeline, completes in under four GPU-hours on a single RTX 4090 plus approximately ten USD of LLM API spend, making the benchmark accessible to academic groups without industrial compute. Extending the suite—adding a method, an additional ground-truth formulation, or a new benchmark axis—is supported by single-file plug-in points documented alongside the harness.

6 EXPERIMENTS

We evaluate DEXPOSURE-FRAMEWORK on the DEXPOSURE-BENCH suite (§5), grouping the six benchmarks under two natural axes: (i) does the forecast–measure pipeline produce accurate, well-calibrated, and timely risk signals (Task I, covering b1, b2, b3, b6), and (ii) do the resulting decision tickets improve upon baselines in precision, stability, and auditability (Task II, covering b4, b5)? The ground truth, dataset split, and reference methods used throughout this section are defined in §5; all results are reported on the frozen test split of Table 5.

6.1 Task I: Network Risk Monitoring

6.1.1 Implementation Details

Data and temporal granularity.: We use the DEX-POSURE dataset [26], comprising weekly directed weighted exposure-graph snapshots sourced from the DeFiLlama API. The dataset spans 2020-03 to 2025-08 (283 weeks), with 50–500 protocol nodes and 100–2,000 exposure edges per snapshot, totalling over 43.7M data entries. Node features include log-TVL, number of token types, maximum token share, Shannon entropy of token distribution, and protocol category (one-hot, 15 classes). Edge weights are log-scaled USD exposure values. We evaluate multi-horizon forecasting on $\mathcal{H} = \{1, 4, 8, 12\}$ weeks.

Temporal split.: We enforce a strict no-leakage temporal split:

Split	Period	Weeks	Usage
Train	2020-03 – 2024-06	~222	FM backbone training
Val	2024-07 – 2024-12	~26	Hyperparameter + threshold tuning
Test	2025-01 – 2025-08	~33	Final evaluation

Stress event windows.: For early-warning evaluation (b2_warning), we define ground-truth stress periods around three major DeFi crises:

Event	Date	Pre-shock (4w)	Event (3w)
Terra/Luna	2022-05-09	03-28 – 05-02	05-02 – 05-23
FTX	2022-11-07	09-26 – 10-31	10-31 – 11-21
SVB/USDC	2023-03-10	02-06 – 03-06	03-06 – 03-27

Forecast interface and predicted-graph construction.: For each epoch t and horizon $h \in \mathcal{H}$, we query the backbone model to obtain a predictive distribution $\mathcal{P}_{t,h}$. We construct an expected-weight predicted graph $\hat{G}_{t+h}$ by thresholding edge existence probabilities at $\pi_{\min}$ and setting $\hat{w}_{pq} := \Pr(e_{pq} = 1) \cdot \mathbb{E}[w_{pq} \mid e_{pq} = 1]$ (Eq. (6)). We set $\pi_{\min} = 0.2$ on the validation split. For uncertainty quantification, we draw $M = 50$ Monte Carlo samples via Bernoulli edge sampling and Gaussian weight perturbation.

Monitoring metrics and alert triggers.: At each horizon, we compute five monitoring functionals $\mathbf{M}_{t,h} = \Phi(\hat{G}_{t+h}, X_t)$ (Table 1): systemic importance (M1), HHI concentration (M3), network density (M4), PageRank Gini (M6), and degree Gini (M7). Baseline bands $(\mu_t^{(j)}, \sigma_t^{(j)})$ are computed via a rolling window of $L = 26$ weeks. An alert fires when a metric exceeds $z = 2$ standard deviations from baseline (Eq. (10)). Alert confidence follows Eq. (11).

Stress scenarios.: We evaluate five standardised scenarios S_0 (Table 2): single protocol failure (S1, 100% shock on top node), bridge cluster failure (S2, 100% on bridges), stablecoin de-peg (S3, 50% on stablecoins), sector-wide lending shock (S4, 30%), and correlated stress (S5, 20% on top-10 nodes). CVaR is computed as the mean of the worst $\lambda = 0.5$ fraction of MC sample losses (Eq. (14)).

Attribution.: For each triggered alert, we rank contributing edges by marginal contribution: $\text{Contrib}(e) := \Phi^{(j)}(\hat{G}_{t+h}) - \Phi^{(j)}(\hat{G}_{t+h} \setminus e)$, retaining the top $K = 10$.

Evaluation protocol.: Figure 3 summarises the evaluation framework. We evaluate Task I along three axes:

- 1) *Risk-metric forecasting (b1_forecast)*: predicted vs. realised metrics using MAE and Spearman rank correlation, for each $h \in \{1, 4, 8, 12\}$.

Metric (Task I)	m5_fm_rules (FM)
b1_forecast RankCorr@h4	0.558
b1_forecast TrendCons@h4	0.628
b1_forecast HHI-MAE@h4 (lower better)	0.032
b2_warning Precision (h1_weighted_degree mean)	0.756
b2_warning F1-warning (h1_weighted_degree mean)	0.007
b2_warning Lead time (h1_weighted_degree weeks)	4.667
b3_calibration ECE (m5_fm_rules only)	0.012
b3_calibration PI coverage (m5_fm_rules only)	0.912
b6_robustness Relative degradation (mean)	0.113

TABLE 8

Task I monitoring quality summary on the 2025 test split.

- 2) *Early-warning quality (b2_warning)*: lead time to stress events, precision/recall under alert budgets $\{5, 10, 20\}$, alert stability (flip rate), and F1-warning.
- 3) *Uncertainty calibration (b3_calibration)*: expected calibration error (ECE), prediction interval coverage (target 90%), PI width, and CRPS. We apply split conformal prediction on the validation set to ensure asymptotically valid coverage.

Reproducibility.: All hyperparameters ($\pi_{\min}$, L , z , K , M , τ_{data}) are selected on the validation split and fixed for test evaluation. Random seeds are fixed for MC sampling; we report mean $\pm$ standard deviation over $R = 3$ independent runs. All benchmark results in this section are pending regeneration after the experiment-audit cleanup. The cleaned pipeline uses fail-closed predictor routing: unavailable FM or EvolveGCN checkpoints now abort the corresponding run rather than falling back to persistence. b2_warning is reported as a shared weighted-degree heuristic (h1_weighted_degree) rather than as a method-specific comparison. Final runtime and hardware details will be filled in after the clean rerun.

6.1.2 Empirical Results

Table 8 and Table 9 are placeholders until the clean rerun completes. The regenerated Task I comparison will report m5_fm_rules (FM+Rules), m1_persistence_rules (Persistence+Rules), h1_weighted_degree for b2_warning, and m3_evolvegcnn only if trained EvolveGCN checkpoints are present.

Artifact	Status
b1_forecast all-horizon forecasting	Pending clean rerun

TABLE 9

b1_forecast all-horizon forecasting results. This table is intentionally blank until new results are generated from the cleaned experiment pipeline.

The previous numerical Task I discussion has been removed because it was tied to stale artifacts, including a m3_evolvegcnn run that matched persistence after metadata removal. We will restore this paragraph only after the new result manifest passes the audit script.

6.2 Task II: Decision-Making

6.2.1 Implementation Details

Decision timeline and inputs.: Task II operates at the same decision epochs as Task I. At each epoch, the decision

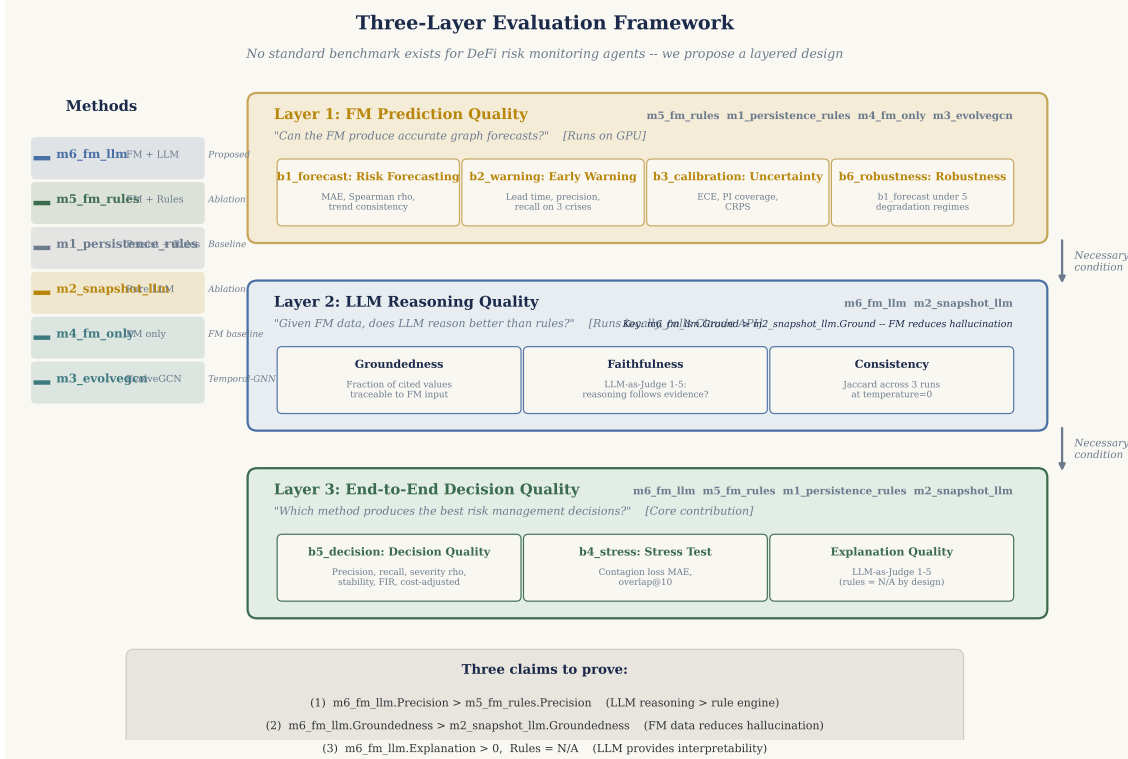


Fig. 3. Evaluation framework: six benchmarks (b1_forecast–b6_robustness) covering risk forecasting, early warning, uncertainty calibration, stress testing, decision quality, and robustness.

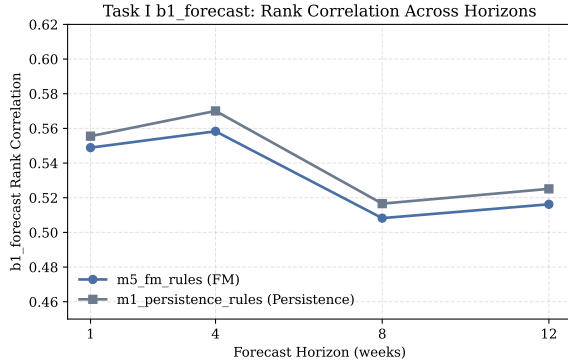


Fig. 4. Task I b1_forecast rank correlation across forecast horizons.

agent receives: (i) the current state (G_t, X_t) , (ii) multi-horizon forecasts $\{\hat{G}_{t+h}\}_{h \in \mathcal{H}}$ and optional MC samples, (iii) monitoring alerts $\mathcal{A}_t$ with evidence and confidence scores, and (iv) the scenario summary $\mathcal{R}_t$.

Action space and playbook. We use a four-level conservative playbook (Table 3): Monitor (severity weight 0.25, always feasible), Investigate (0.50, always feasible), Recommend-Reduce (0.75, requires SAFE_MODE = 0 and ≥ 2 alerts), and Contingency (1.00, requires SAFE_MODE = 0 and ≥ 3 alerts). Intervention-type actions additionally require mean alert confidence $\bar{C}_t \geq \tau_{\text{conf}} = 0.6$.

Ticket scoring. Tickets are ranked by $\text{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \text{Imp}_t$ (Eq. (16)), where w_u is the playbook severity weight, $\bar{C}_t$ is mean alert confidence, and Imp_t is the mean CVaR across all scenarios (defaulting to 1.0 when no scenar-

ios are available). We output the top- $N = 10$ tickets.

Target selection. For intervention tickets, targets are selected from the union of top- K attributed nodes across triggering alerts and worst-case scenarios, ranked by frequency-weighted vote.

Evaluation protocol. We evaluate Task II using both offline and scenario-based criteria:

- 1) *Offline (b5_decision):* ticket precision (fraction followed by real stress), target stability (Jaccard between adjacent epochs), audit completeness (evidence bundle coverage), suppression rate (safe-mode activation), and false intervention rate.
- 2) *Scenario-based (b4_stress):* stress-test contagion prediction accuracy—loss MAE, distressed count MAE, propagation depth MAE, and target overlap@ K between predicted and actual networks under scenarios S1–S5.

Reproducibility. All decision hyperparameters (K , N , τ_{data} , τ_{conf} , λ) are tuned on the validation split and fixed for testing. MC components use fixed seeds; we report mean $\pm$ standard deviation over $R = 3$ runs.

6.2.2 Empirical Results

Table 10 is pending clean regeneration. The rerun will compare m1_persistence_rules (Persistence+Rules), m5_fm_rules (FM+Rules), m2_snapshot_llm (CurrentSnapshot+LLM), m6_fm_llm (FM+LLM), and m7_fm_llm_gated (FM+LLM+RulesGate). The gated LLM variant is included to separate FM-conditioned target selection from unconstrained LLM intervention severity.

Figure 5 visualises the key metrics (precision, stability, and cost-adjusted score).

Method		Prec	Recall	Stabil	SevRisk	FIR	Ground	Consist	Explains	CostAdj
m1_persistence_rules	Persist+Rules	0.720	0.004	0.514	N/A	0.000	N/A	1.000	Pending clean rerun	0.720
m5_fm_rules	FM+Rules	0.600	0.010	0.257	N/A	0.000	N/A	1.000	N/A	0.600
m2_snapshot_llm	Pure LLM	0.575	0.009	0.532	N/A	0.000	N/A	1.000	N/A	0.575
m6_fm_llm	FM+LLM	0.570	0.012	0.488	N/A	0.000	N/A	1.000	N/A	0.570
m7_fm_llm_gated	FM+LLM+RulesGate	0.580	0.012	0.435	-0.003	0.437	1.000	0.855	2.7/5	0.327

TABLE 10
Task II method comparison (Table 3).

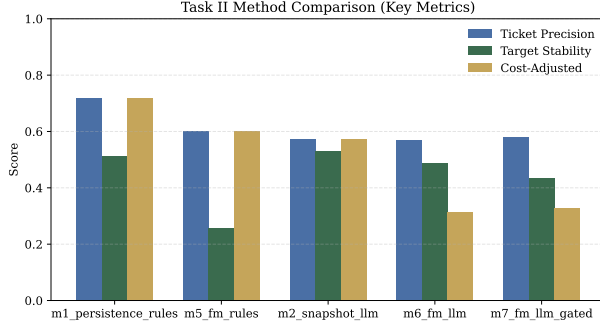


Fig. 5. Task II method comparison on key decision metrics.

The previous Task II numerical discussion has been removed because the LLM checkpoint files and raw assessments disagreed, and because the ungated FM+LLM policy was not comparable to the conservative rule engine. We will restore this paragraph after the clean LLM and rule-based reruns pass the audit script.

Artifact	Status
b5_decision crisis-period decision quality	Pending clean rerun

TABLE 11

b5_decision crisis-period decision quality. This table is intentionally blank until new results are generated from the cleaned experiment pipeline.

Early-warning timeline analysis.: To determine whether FM alerts constitute genuine *early warnings* or merely post-hoc damage reports, we conduct a per-week alert timeline analysis around the Terra/Luna (2022-05-09) and FTX (2022-11-07) events. Table 12 summarises the results.

Artifact	Status
Early-warning timeline	Pending clean rerun

TABLE 12

Early-warning timeline. This table is intentionally blank until new results are generated from the cleaned experiment pipeline.

The early-warning timeline and b4_stress stress-test discussion are also pending clean regeneration. We keep the table hooks in place so the new artifacts can be inserted directly after rerun.

6.3 Ablation Study

We validate the necessity of key design components through ablation experiments under both standard and stress conditions (Table 14).

Standard conditions (Panel A): Under clean 2025 test data, the confidence gate (A2) proves essential: removing it increases the false intervention rate from 0.000 to 0.400. The scenario engine (A3) is structurally necessary. Data-health gating (A1) and multi-horizon diversity (A6) show no measurable impact because data quality is high and $h = 4$ dominates alert generation.

Stress conditions (Panels B–C): To test whether A1 and A6 are genuinely unnecessary or merely dormant, we evaluate them under adversarial conditions. For A1, we apply severe data degradation (80–95% of features and edges removed, reducing DH_t to 0.59–0.75) and isolate the data-health gate from the confidence gate by setting $\tau_{\text{conf}} = 0$. Under these conditions, disabling the data-health gate increases the false intervention rate from 0% to 41–48% (Panel B), confirming that A1 functions as a *safety mechanism that activates under data degradation*. For A6, we compare single-horizon ($h = 4$) against multi-horizon ($h \in \{1, 4, 8, 12\}$) monitoring during the Terra/Luna crisis period: multi-horizon generates $5\times$ more alerts while maintaining identical precision (Panel C), confirming that A6 provides broader risk coverage when network dynamics are rapid. These results demonstrate that all four components are necessary for reliable operation: A2 and A3 are active in all conditions, while A1 and A6 are safety reserves that engage under data degradation and crisis dynamics, respectively.

7 CONCLUSION

DeFi has grown into a tightly coupled financial ecosystem where tokenised liabilities link protocols into a dynamic web of credit exposures, so fast and reliable *risk monitoring* and *decision recommendation* are increasingly critical for investors, protocol teams, and regulators. Although recent “agentic” workflows aim to automate surveillance and intervention planning, they are often not yet dependable in real deployments, largely because they lack a trustworthy *foundation model backbone* capable of forecasting the evolving exposure network with strong data efficiency and well-calibrated uncertainty. Leveraging DeXposure-FM, a graph time-series foundation model trained on more than 43.7 million data entries, we introduce *DeXposure-Agent*, which to the best of our knowledge is the first agentic system that integrates a graph time-series foundation model backbone for DeFi exposure-network risk monitoring and decision recommendation. DeXposure-Agent continuously ingests exposure snapshots, uses DeXposure-FM to predict future credit-exposure graphs, derives systemic risk indicators and standardised stress-test scenarios, and produces auditable alerts together with conservative, playbook-driven recommendations governed by explicit data-health and uncertainty gates. Across six benchmarks and stress-test settings, including low-data and weakly supervised

Artifact	Status
Ablation study	Pending clean rerun

TABLE 14

Ablation study. This table is intentionally blank until new results are generated from the cleaned experiment pipeline.

regimes, DeXposure-Agent (1) delivers earlier and more stable early-warning signals of systemic fragility, (2) improves risk-metric forecasting and trend detection beyond naïve baselines, and (3) generates higher-quality recommendation tickets with improved grounding and consistency when combining the foundation model backbone with LLM-based reasoning.

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